

Лабораторная работа

Номер 1

Приходько И. И.

4 сентября 2026

Российский университет дружбы народов, Москва, Россия

Информация

- Приходько Иван Иванович
- Студент
- Российский университет дружбы народов

Приобретение практических навыков для установки операционной системы на виртуальную машину, настройки минимально необходимых для дальнейшей работы сервисов.

Для начала добавляем установочный диск

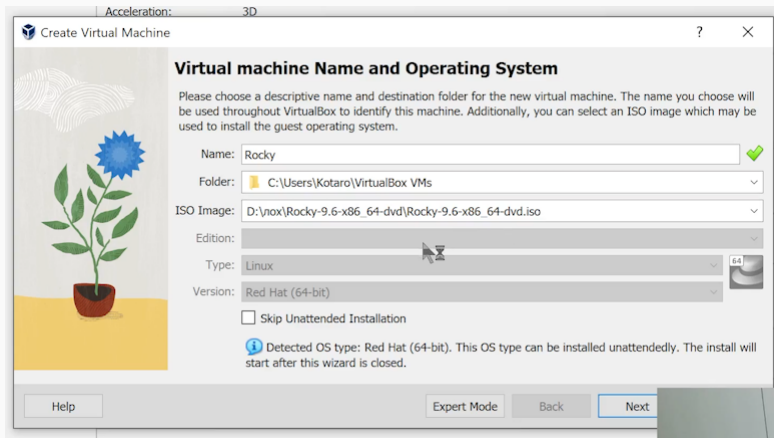


Рис. 1: Установочный диск

Устанавливаем имя и пароль

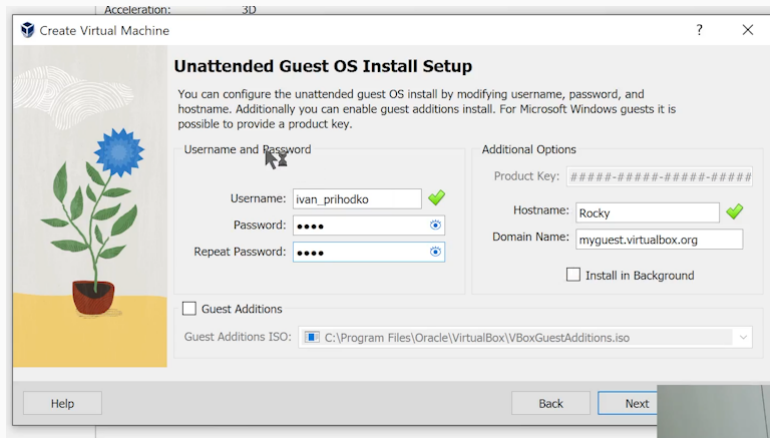


Рис. 2: Установка логина и пароля

Устанавливаем процессоры и оперативную память

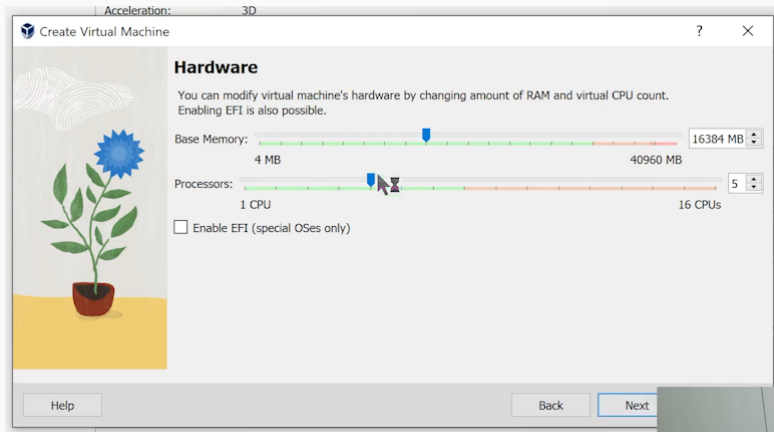


Рис. 3: Установка процессоров и оперативную память

Устанавливаем размер жесткого диска

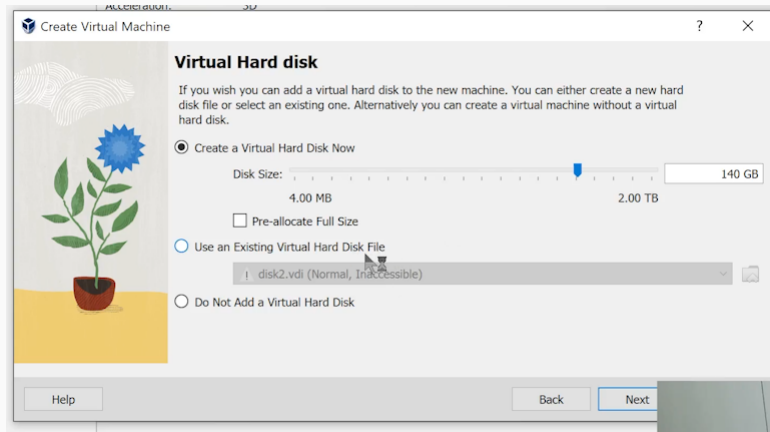


Рис. 4: Установка размера жесткого диска

После этого устанавливаем язык

WELCOME TO ROCKY LINUX 9.6.

What language would you like to use during the installation process?

Русский	<i>Russian</i>
العربية	<i>Arabic</i>
English	<i>English</i> >
Français	<i>French</i>
Deutsch	<i>German</i>
日本語	<i>Japanese</i>
中文	<i>Mandarin Chinese</i>
Español	<i>Spanish</i>
Afrikaans	<i>Afrikaans</i>
አማርኛ	<i>Amharic</i>
অসমীয়া	<i>Assamese</i>
Asturianu	<i>Asturian</i>
Беларуская	<i>Belarusian</i>
Български	<i>Bulgarian</i>

English (United States)

English (United Kingdom)
English (India)
English (Australia)
English (Canada)
English (Denmark)
English (Ireland)
English (New Zealand)
English (Nigeria)
English (Hong Kong SAR China)
English (Philippines)
English (Singapore)
English (South Africa)
English (Zambia)
English (Zimbabwe)
English (Botswana)

Устанавливаем пароль для админа

The root account is used for administering the system. Enter a password for the root user.

Root Password: 

Empty

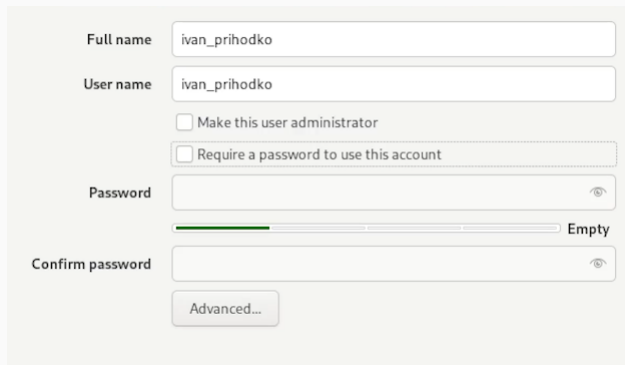
Confirm: 

☐ Lock root account

☐ Allow root SSH login with password

Рис. 6: Установка пароля для админа

Установка логина и пароля устройства



The screenshot shows the user creation interface during the Kali Linux installation. It includes the following fields and options:

- Full name:** A text input field containing the value "ivan_prihodko".
- User name:** A text input field containing the value "ivan_prihodko".
- Make this user administrator:** An unchecked checkbox.
- Require a password to use this account:** An unchecked checkbox.
- Password:** A text input field that is currently empty. To its right is a strength indicator bar with a green segment on the left and the word "Empty" at the end.
- Confirm password:** A text input field, currently empty, with a toggle icon on the right.
- Advanced...:** A button located below the password fields.

Рис. 7: Установка логина и пароля для устройства

Добавляем Debugging Tools

SOFTWARE SELECTION

Done

ROCKY LINUX 9.6 INSTALLATION

 us

Help!

Base Environment

- ☒ **Server with GUI**
An integrated, easy-to-manage server with a graphical interface.
- ☐ **Server**
An integrated, easy-to-manage server.
- ☐ **Minimal Install**
Basic functionality.
- ☐ **Workstation**
Workstation is a user-friendly desktop system for laptops and PCs.
- ☐ **Custom Operating System**
Basic building block for a custom Rocky Linux system.
- ☐ **Virtualization Host**
Minimal virtualization host.

Additional software for Selected Environment

- ☒ **Debugging Tools**
Tools for debugging misbehaving applications and diagnosing performance problems.
- ☐ **DNS Name Server**
This package group allows you to run a DNS name server (BIND) on the system.
- ☐ **File and Storage Server**
CIFS, SMB, NFS, iSCSI, iSER, and iSNS network storage server.
- ☐ **FTP Server**
These tools allow you to run an FTP server on the system.
- ☐ **Guest Agents**
Agents used when running under a hypervisor.
- ☐ **Infiniband Support**
Software designed for supporting clustering, grid connectivity, and low-latency, high bandwidth storage using RDMA-based InfiniBand, iWARP, RoCE, and OPA fabrics.
- ☐ **Mail Server**
These packages allow you to configure an IMAP or SMTP mail server.
- ☐ **Network File System Client**
Enables the system to attach to network storage.
- ☐ **Network Servers**
These packages include network-based servers such as DHCP, Kerberos and NIS.
- ☐ **Performance Tools**
Tools for diagnosing system and application-level performance problems.
- ☐ **Remote Desktop Clients**

Устанавливаем жесткий диск (рис. (fig:009?)).

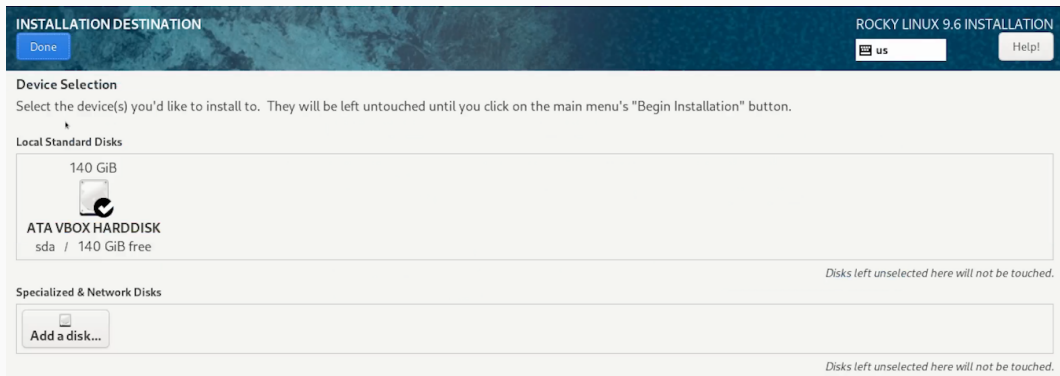
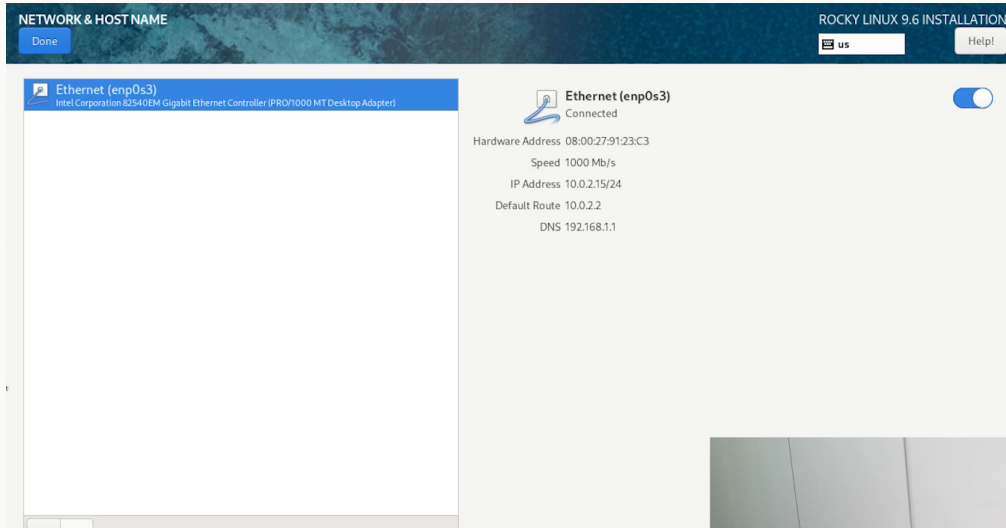


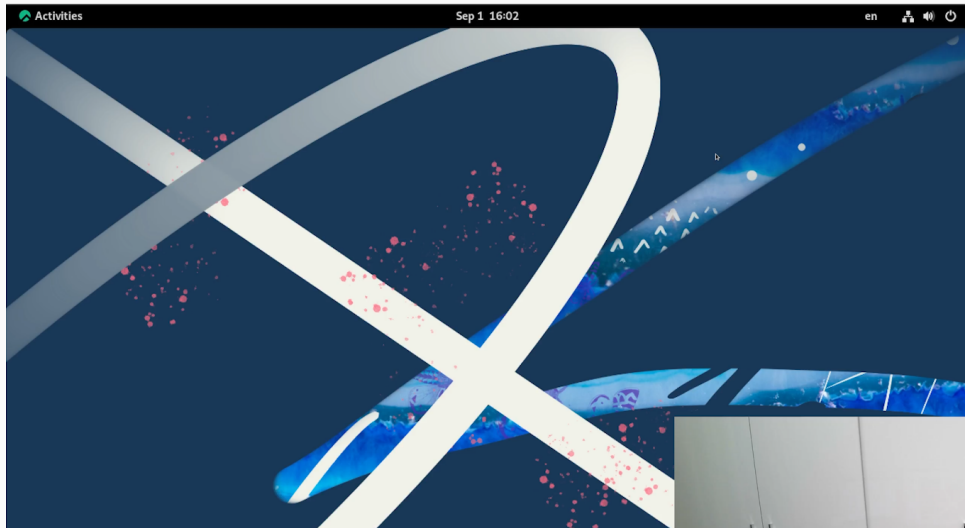
Рис. 9: Установка жесткого диска

Вписываем имя хоста (рис. (fig:010?)).



Установка Kali Linux

После этого ждем установку пакетов



Находим: 1. Версия ядра Linux (Linux version). 2. Частота процессора (Detected Mhz processor). 3. Модель процессора (CPU0). 4. Объем доступной оперативной памяти (Memory available). 5. Тип обнаруженного гипервизора (Hypervisor detected). 6. Тип файловой системы корневого раздела. 7. Последовательность монтирования файловых систем.


```
ournal.  
[ 7.173668] piix4_smbus 0000:00:07.0: SMBus Host Controller at 0x4100, revision 0  
[ 7.186414] input: PC Speaker as /devices/platform/pcspkr/input/input6  
[ 7.257602] intel_rapl_common: Found RAPL domain package  
[ 7.257612] intel_rapl_common: Found RAPL domain core  
[ 7.597243] snd_intel8x0 0000:00:05.0: allow list rate for 1028:0177 is 48000  
[ 7.655071] XFS (sda1): Mounting V5 Filesystem ccb46ald-0f63-4670-bd18-9e4db14c960c  
[ 7.655494] XFS (dm-2): Mounting V5 Filesystem 1110cd02-1dcb-4d45-9933-2d7edc91d0c5  
[ 7.695701] XFS (sda1): Ending clean mount  
[ 7.742537] XFS (dm-2): Starting recovery (logdev: internal)  
[ 7.825286] XFS (dm-2): Ending recovery (logdev: internal)  
[ 8.548740] NET: Registered PF_QIPCRTR protocol family  
[ 9.083699] Warning: Unmaintained driver is detected: ip_set  
[ 9.162054] e1000: enp0s3 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX  
[ 12.021650] rfkill: input handler disabled  
[ 16.781587] rfkill: input handler enabled  
[ 18.212220] rfkill: input handler disabled  
[ 31.457346] clocksource: Long readout interval, skipping watchdog check  
sec: 1220846082 wd_nsec: 1220846317  
[root@ivanprihodko ~]#
```

Рис. 12: Поиск информации о компьютере

```
[ 0.000000] kvm-clock: Using msrs 4b564d01 and 4b564d00
[ 0.000004] kvm-clock: using sched offset of 9381268040 cycles
[ 0.000008] clocksource: kvm-clock: mask: 0xffffffffffffffff max_cycles: 0x1cd42e4dffb, max_idle_ns: 8
81590591483 ns
[ 0.000013] tsc: Detected 3792.870 MHz processor
[ 0.001270] e820: update [mem 0x00000000-0x00000fff] usable ==> reserved
[ 0.001274] e820: remove [mem 0x000a0000-0x000fffff] usable
[ 0.001279] last_pfn = 0x420000 max_arch_pfn = 0x400000000
```

Рис. 13: Поиск информации о компьютере

```
[ivan_prihodko@ivanprihodko ~]$ dmesg | grep -i "cpu"
[ 0.001641] ACPI: SSDT 0x00000000DFFF02C0 00036C (v01 VBOX VBOXCPU 00000002 INTL 20100528)
[ 0.009131] CPU topo: Max. logical packages: 1
[ 0.009132] CPU topo: Max. logical dies: 1
[ 0.009132] CPU topo: Max. dies per package: 1
[ 0.009134] CPU topo: Max. threads per core: 1
[ 0.009135] CPU topo: Num. cores per package: 6
[ 0.009135] CPU topo: Num. threads per package: 6
[ 0.009136] CPU topo: Allowing 6 present CPUs plus 0 hotplug CPUs
[ 0.012282] setup_percpu: NR_CPUS:8192 nr_cpumask_bits:6 nr_cpu_ids:6 nr_node_ids:1
[ 0.012557] percpu: Embedded 64 pages/cpu s225280 r8192 d28672 u262144
[ 0.012561] pcpu-alloc: s225280 r8192 d28672 u262144 alloc=1*2097152
[ 0.012563] pcpu-alloc: [0] 0 1 2 3 4 5 - -
[ 0.035785] SLUB: HWalign=64, Order=0-3, MinObjects=0, CPUs=6, Nodes=1
[ 0.045722] rcu: RCU restricting CPUs from NR_CPUS=8192 to nr_cpu_ids=6.
[ 0.045725] rcu: Adjusting geometry for rcu_fanout_leaf=16, nr_cpu_ids=6
[ 0.165832] smpboot: CPU0: AMD Ryzen 7 5800X 8-Core Processor (family: 0x19, model: 0x21, stepping: 0x0)
[ 0.168962] smp: Bringing up secondary CPUs ...
[ 0.169083] .... node #0, CPUs: #1 #2 #3 #4 #5
[ 0.177586] smp: Brought up 1 node, 6 CPUs
[ 0.199766] cpuidle: using governor menu
[ 0.201871] cryptd: max_cpu_qlen set to 1000
[ 0.203966] ACPI: _OSC evaluation for CPUs failed, trying _PDC
[ivan_prihodko@ivanprihodko ~]$
```

Рис. 14: Поиск информации о компьютере

```
[ 0.009172] PM: hibernation: Registered nosave memory: [mem 0xffffc0000-0xffffffff]  
[ 0.035663] Memory: 3674072K/16776760K available (16384K kernel code, 5766K rwdata, 13624K rodata, 4048K init)
```

Рис. 15: Поиск информации о компьютере

```
[ivan_prihodko@ivanprihodko ~]$ dmesg | grep -i "hypervisor"
[ 0.000000] Hypervisor detected: KVM
[ 3.386482] vmwgfx 0000:00:02.0: [drm] *ERROR* vmwgfx seems to be running on an unsupported hypervisor.
[ivan_prihodko@ivanprihodko ~]$ dmesg | grep -i "filesystem"
[ivan_prihodko@ivanprihodko ~]$ dmesg | grep -i "filesystem"
[ 4.740462] XFS (dm-0): Mounting V5 Filesystem 4db8c666-813e-4981-a008-06df75a7a52a
[ 7.655071] XFS (sda1): Mounting V5 Filesystem ccb46a1d-0f63-4670-bd18-9e4db14c960c
[ 7.655494] XFS (dm-2): Mounting V5 Filesystem 1110cd02-1dcb-4d45-9933-2d7edc91d0c5
[ivan_prihodko@ivanprihodko ~]$
```

Рис. 16: Поиск информации о компьютере

В ходе данной лабораторной работы были получены знания для установки Rocky.